Rasmus Frigaard Lemvig - Curriculum Vitae

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Education

University of Copenhagen
PhD student. Principal supervisor: Martin Bladt. Co-supervisor: Mogens Steffensen *(May 2025 - present)*
Working thesis title: *Multi-state models: modern methods, machine learning and meditations* $((MM)^3)$

University of Copenhagen
Master of Science in Actuarial Mathematics *(Sep 2023 - Jan 2025)*
Thesis: Machine learning methods for survival and multi-state models
Two-part thesis on reserving under stress scenarios in life insurance and insurance pricing uisng GLMs and machine learning.

University of Copenhagen
Bachelor of Science in Actuarial Mathematics *(Sep 2021 - Jun 2023)*

University of Copenhagen
Bachelor of Science in Mathematics *(Sep 2018 - Jun 2021)*
Thesis: *Cubic and Quartic Reciprocity*

Experience

University of Copenhagen - Department of Mathematical Sciences
PhD student *(May 2025 - present)*. PhD student in the section of Insurance and Economics (IE). Interests include the mathematics and statistics of life insurance, in particular multi-state modelling, and their intersection with machine learning. Other interests include mathematical aspects of risk management and finance.

Scientific assistant *(Feb 2025 - May 2025)*

Teaching assistant *(Sep 2020 - present)* 24 times in total, spanning 20 different courses. Noteworthy courses include:

- Statistics for Insurance
- Quantitative Risk Management
- Topics in Probability
- Basic Life Insurance Mathematics
- Statistical Methods
- Brownian Motion
- Continuous Time Finance
- Advanced Probability Theory 2
- Stochastic Processes in Non-Life Insurance
- Insurance and Law
- Probability Theory 2

- Probability Theory

My complete teaching history can be found here: https://rasmusfl.github.io/work.html.

Manuscripts and projects

1. Random Jump Forests
   Martin Bladt and Rasmus Frigaard Lemvig
2. Local estimation of transition rates of jump processes through discretization
   Martin Bladt and Rasmus Frigaard Lemvig
3. *Consistency of Honest Decision Trees and Random Forests*
   Martin Bladt and Rasmus Frigaard Lemvig
4. *JumpForests: A random forest R package for jump processes*
   Martin Bladt and Rasmus Frigaard Lemvig
5. Lecture notes for the course *Quantitative Risk Management (QRM)*
   Rasmus Frigaard Lemvig
6. Lecture notes for the course *Stochastic Processes in Non-Life Insurance (SkadeStok) 2023/2024*
   Rasmus Frigaard Lemvig

Programming proficiencies

Experience with programming in **R, Julia, Python/Cython and C++**, including R package development with **RCpp**.

Volunteer work

Danish Youth Association of Science (UNF)
Volunteer *(Jan 2016 - present).* Some highlights include:

- **Cofounder of, coordinator and teacher at** UNF Science Weekend (2019 - 2023), a biannual weekend camp for elementary school students with 85 participants and 30 volunteers.
- **Teacher at** Mathematics Camp from 2021 to 2025 and at Machine Learning Camp 2024.
- **Teacher at** Week in mathematical modelling 2025, a camp with focus on differential equations and their applications. Teacher and coordinator in 2026.
- **Teacher** in several workshops. Topics: Linear algebra, calculus, number theory, graph theory, sums and sequences, differential equations.
- **Coordinator of and teacher in** the project Evenings of Natural Science Education where we taught young Ukrainian refugees.

A detailed description of my work for UNF can be found here: https://rasmusfl.github.io/volunteer.html.

University of Copenhagen - Department of Mathematical Sciences
Student representative in the Board of Education *(Jun 2022 - Jan 2025)*